WSABI Summer Research Lab **Summer 2026**

Lecture 7: Causal Inference

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7.1 Introduction

Throughout this course so far, we have mostly studied patterns in observed data. We have estimated conditional means, fit regression models, measured associations, and asked whether apparent patterns are large relative to chance variation. Those tools are essential in sports analytics: they help us describe what happened, forecast what might happen next, and summarize how players or teams tend to perform in particular situations.

This lecture changes the target. Instead of asking only what tends to happen, we ask what would happen if we changed a decision.

> **Causal question.** If we intervened and changed one decision, while holding the relevant background conditions fixed, how would the outcome change?

Our running example will be NBA timeout decisions after a 7-point drop in score margin over the last 3 minutes. A predictive model might show that teams often perform worse after calling timeout than after letting play continue. That association may be useful for forecasting. It does not, by itself, answer the coaching question: should the coach call timeout now?

The difficulty is that coaches do not call timeouts at random. They call them in particular situations: when the score swing has become dangerous, when a lineup looks disorganized, when the crowd is involved, when a star needs rest, or when a tactical mismatch feels urgent. A comparison between moments where coaches called timeout and moments where they did not is therefore not automatically a comparison between otherwise similar situations.

This distinction matters because prediction and decision-making have different targets. If our only goal is to forecast what usually happens after a timeout decision, a stable association may be enough. If our goal is to recommend an action, we need a causal comparison. Association compares moments as they appeared in the data. Causation asks how the outcome would change under a different decision in the same situation.

That is why causal inference is hard. The comparison we want is partly counterfactual. For any actual timeout decision, we observe one choice and one outcome. We do not also observe what would have happened if the coach had made the other choice in the same moment. Causal inference is the discipline of reasoning carefully about that missing comparison.

The lecture proceeds in four steps. First, we define the timeout decision problem and explain why the raw association is not a causal effect. Second, we introduce target-trial thinking and potential outcomes, which make the desired comparison explicit. Third, we state the main assumptions needed to interpret observational comparisons causally. Finally, we connect those assumptions to common methods such as regression adjustment, matching, propensity scores, weighting, and sensitivity analysis.

7.2 Running Example: NBA Timeouts After a 7-Point Drop

The question throughout the lecture is: when a team's score margin has worsened by at least 7 points over the last 3 minutes of game time, what is the causal effect of calling a timeout now instead of letting play continue?

Timeout decisions make a useful causal example because the intervention is real, the counterfactual is intuitive, and the confounding is obvious. A coach watches the opponent score in a way that pushes the team's score-margin change over the previous 180 seconds to -7 or worse. The coach either calls timeout now or lets play continue. We observe the result under the choice actually made, but we do not observe what would have happened if the same team, in the same game state, had made the other choice.

For now, restrict attention to moments where the realistic alternatives are "call timeout now" and "do not call timeout now." Let

$$\begin{aligned} Z_i = 1 &\quad \text{coach calls timeout immediately,} \\ Z_i = 0 &\quad \text{coach lets play continue without a timeout.} \end{aligned}$$

The unit i is a timeout decision opportunity immediately after an opponent basket or free throw, at the first moment in a timeout-free stretch when the team's score margin over the previous 180 seconds reaches -7 or worse. The outcome Y_i could be defined in several ways:

- change in score margin over the next 4 possessions;
- opponent points over the next 2 possessions;
- team net score over the next 3 minutes of game time;
- whether the negative momentum reverses within the next possession or two.

Different outcomes answer different causal questions. In these notes, we will usually speak as if Y_i is the change in score margin for the team after treatment time over the next 3 minutes of game time. This outcome stays close to the intervention while preserving a direct basketball interpretation.

7.2.1 Raw Association Is Not Enough

The most intuitive comparison to make is

$$\mathbb{E}[Y_i \mid Z_i = 1] - \mathbb{E}[Y_i \mid Z_i = 0].$$

This compares moments where coaches called timeout to moments where they did not. Why is this not automatically a causal effect?

The problem is selection into treatment. Coaches call timeouts more often in some situations than in others:

- when the short-term score swing is already severe;
- when the score margin and time remaining make the moment feel urgent;
- when the lineup looks tired, disorganized, or mismatched;
- when the coach is generally more or less timeout-prone;

- when private information, such as a player's fatigue or a tactical concern, changes the decision.

The naive comparison therefore mixes two things: the effect of the decision itself, and the pre-treatment factors that made the coach more or less likely to choose that decision in the first place. That mixing is called *confounding*. A confounder is a variable that affects both treatment assignment and the outcome.

7.3 A Graphical View of Confounding

A simple graph makes the design problem concrete:

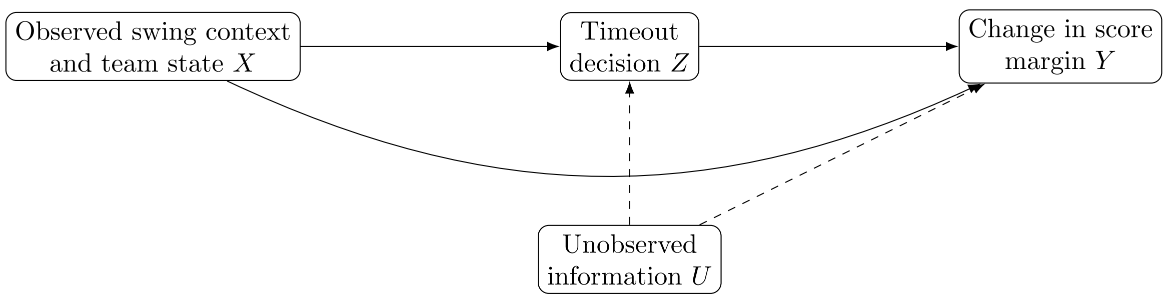

Figure 7.1: Confounding in the timeout example. Pre-treatment variables affect both the decision and the outcome. Unobserved information may do the same.

The idea is simple: the same pre-treatment variables can affect both the decision and the outcome. Recent score swing, current score margin, time remaining, lineup quality, team strength, and coaching tendencies all help determine whether a coach calls timeout. They also help determine what happens next. If these variables differ across the timeout and no-timeout groups, the groups are not automatically comparable.

7.4 Designing Our Ideal Comparison

The cleanest way to answer the timeout question would be a randomized experiment.

Imagine that, for all eligible timeout opportunities, the league randomly assigned the decision:

$$\begin{aligned} Z_i &= 1 \quad \text{call timeout now,} \\ Z_i &= 0 \quad \text{let play continue.} \end{aligned}$$

Then, up to chance variation, the timeout and no-timeout moments would have similar recent score swings, score margins, periods, teams, opponents, lineups, and coaches. The experimental logic is simple: if the only systematic difference between two groups is the treatment, then a systematic difference in outcomes is evidence about the treatment effect.

Randomization is powerful because it creates comparability. It balances not only measured variables, such as recent score swing and score margin, but also unmeasured variables, such as coach frustration, private tactical concerns, or knowledge that a player is exhausted.

Of course, the NBA is not going to randomize timeout decisions for our research project. In practice, we usually have observational data: coaches made their own decisions, and we try to learn from those decisions after the fact. Observational causal inference tries to reconstruct, as well as possible, the controlled comparison that a randomized experiment would have created.

Before writing down a model, it helps to make the hypothetical experiment explicit. In causal inference, this idealized design is often called the *target trial*.

Design element	Timeout decision example
Units	Eligible timeout opportunities immediately after an opponent scoring play, at the first threshold crossing in a timeout-free stretch.
Treatment	Coach calls timeout now.
Control	Coach lets play continue without a timeout.
Time zero	The dead-ball moment immediately after the opponent scores and before the next live possession.
Outcome	Change in score margin over the next 3 minutes of game time.
Eligibility	Timeout-free moments where the score-margin change over the previous 180 game-seconds is -7 or worse, with enough time remaining to measure the local outcome and overlap in observed game states.
Covariates	Recent score swing, current score margin, period, time remaining, home/away, timeouts remaining if available, team strength, opponent strength, lineup quality, fatigue, and foul trouble if available.
Estimand	The average effect among all eligible threshold-crossing timeout opportunities, or the average effect among the moments where coaches actually called timeout.
Main threat	Coaches choose strategically using tactical information that may be missing from the data.

Table 7.1: Target-trial thinking for an observational study of NBA timeout decisions.

This design-first view keeps the analysis grounded. The observational data are useful only to the extent that they help us approximate this missing experiment.

7.4.1 The Best Comparison We Can Usually Get

In observational data, the best comparison we can usually hope for is one that compares moments that are nearly identical in their *measured covariates* before the decision:

- similar recent score swing;
- similar current score margin;
- similar time remaining and period;
- similar home/away status;
- similar offensive and defensive team strength;
- similar lineup and fatigue context, if those variables are observed.

Then the treated and control groups differ mainly in the decision. That is about as close as observational data can get to the controlled comparison we actually want.

7.5 Potential Outcomes

Target-trial thinking gives us the design. Potential outcomes give us a way to name the two outcomes we want to compare for the same unit: the outcome if the coach calls timeout now, and the outcome if the coach does not.

Definition 7.1 (Potential Outcomes)**.** *For unit i, let*

$$\begin{aligned} Y_i(1) &= \textit{the outcome unit } i \textit{ would have if the coach called timeout now}, \\ Y_i(0) &= \textit{the outcome unit } i \textit{ would have if the coach did not call timeout now.} \end{aligned}$$

These are called the potential outcomes*.*

In the target-trial language, $Y_i(1)$ and $Y_i(0)$ are the two outcomes the same timeout opportunity would produce under the two treatment arms. For a particular late-second-quarter score that pushes the team to -7 over the previous 180 seconds, $Y_i(1)$ is the change in score margin over the next 3 minutes if the coach calls timeout now, and $Y_i(0)$ is the change in score margin over the next 3 minutes if the coach lets play continue. The individual causal effect is

$$\tau_i = Y_i(1) - Y_i(0).$$

This expression is conceptually simple but empirically impossible to observe. In a real game, the coach makes one decision. We do not get to rewind the game, restore the same players, fatigue, defensive alignment, crowd, and clock state, and then observe the other decision.

> **Fundamental problem of causal inference.** For each unit, we observe at most one potential outcome. The other potential outcome is counterfactual.

If Z_i denotes the treatment actually received, then the observed outcome is

$$Y_i^{\text{obs}} = Z_i Y_i(1) + (1 - Z_i) Y_i(0).$$

The missing potential outcome is what makes causal inference different from ordinary prediction.

7.5.1 Causal Effects Are Always Comparative

A causal effect is always defined relative to some alternative. So the question "Does calling timeout help?" is incomplete unless we define what the alternative is.

In our timeout example, the comparison is calling timeout now versus letting play continue. If the realistic alternatives are calling timeout now, waiting one more possession, or substituting without a timeout, then we need either a multivalued treatment or a clearly stated pairwise comparison. The estimand changes when the comparison changes.

If the comparison is not clearly defined, then neither is the causal estimand.

7.6 What Are We Trying to Estimate?

Because individual causal effects are usually unobservable, we typically target average causal effects.

Definition 7.2 (Average Treatment Effect)**.** *The average treatment effect is*

$$\mathrm{ATE} = \mathbb{E}[Y_i(1) - Y_i(0)] = \mathbb{E}[Y_i(1)] - \mathbb{E}[Y_i(0)].$$

It is the average effect if every eligible threshold-crossing timeout opportunity were handled by calling timeout rather than if every eligible threshold-crossing timeout opportunity were handled by letting play continue.

Definition 7.3 (Average Treatment Effect on the Treated)**.** *The average treatment effect on the treated is*

$$\mathrm{ATT} = \mathbb{E}[Y_i(1) - Y_i(0) \mid Z_i = 1].$$

It is the average effect for the threshold-crossing timeout opportunities where coaches actually called timeout.

The distinction matters. The ATE asks about all eligible threshold-crossing timeout opportunities. The ATT asks about the subset of moments in which coaches actually chose to call timeout. If coaches tend to call timeout only when the swing looks especially dangerous even within this thresholded set, the ATT may be less favorable than the ATE.

These estimands are averages, not one-play effects. They are specific to the target population we define, such as the set of eligible threshold-crossing timeout opportunities with plausible overlap between timeout and no-timeout decisions. If we wanted the effect for a narrower game state, such as a late-third-quarter -7 swing with a small deficit, that would be a more local conditional effect rather than the overall ATE or ATT.

7.6.1 The Association Is Not the Estimand

We can now state more precisely why the naive comparison is not the causal quantity we want.

The raw difference in observed outcomes is

$$\mathbb{E}[Y_i^{\mathrm{obs}} \mid Z_i = 1] - \mathbb{E}[Y_i^{\mathrm{obs}} \mid Z_i = 0].$$

Using the observed-outcome relationship, this equals

$$\mathbb{E}[Y_i(1) \mid Z_i = 1] - \mathbb{E}[Y_i(0) \mid Z_i = 0].$$

The ATE is

$$\mathbb{E}[Y_i(1)] - \mathbb{E}[Y_i(0)].$$

These are not the same quantity. The association compares one group of moments that received a timeout to a different group of moments that did not. The causal estimand compares the same target population under treatment to that same target population under control.

A useful next step is to rewrite the observed comparison in a way that separates the causal part from the selection part. The algebra is simple: we rewrite the observed outcomes in terms of potential outcomes, then add and subtract the same term so that the expression breaks into two pieces. This gives

$$\begin{aligned}
&\mathbb{E}[Y_i^{\mathrm{obs}} \mid Z_i = 1] - \mathbb{E}[Y_i^{\mathrm{obs}} \mid Z_i = 0] \\
&\quad = \underbrace{\mathbb{E}[Y_i(1) - Y_i(0) \mid Z_i = 1]}_{\text{ATT}} + \underbrace{\{\mathbb{E}[Y_i(0) \mid Z_i = 1] - \mathbb{E}[Y_i(0) \mid Z_i = 0]\}}_{\text{selection bias}}.
\end{aligned}$$

For timeout decisions, selection bias is easy to imagine: the situations where coaches call timeout may already be situations with worse counterfactual outcomes under no timeout than the situations where coaches let play continue.

7.7 The Assumptions Behind a Causal Interpretation

Once the design and estimand are clear, the next question is what assumptions let us use observational data as a substitute for the experiment we did not get to run. In the target-trial picture, these assumptions are the bridge from observed associations to counterfactual causal claims. They tell us when an adjusted comparison can be interpreted as a causal comparison.

7.7.1 Assumption 1: Consistency and Well-Defined Treatments

Definition 7.4 (Consistency)**.** *If unit i receives treatment level $Z_i = z$, then the observed outcome equals the potential outcome under that treatment:*

$$Y_i^{\text{obs}} = Y_i(z).$$

This can sound automatic at first, but it is not. The statement only makes sense if the treatment itself is well-defined. In plain language, if we say a team "received treatment $Z = 1$," we need to know what that treatment actually means before we can say that the observed outcome equals the corresponding potential outcome.

For timeouts, "call timeout now" could mean a full timeout, a short timeout, a timeout that produces a substitution, or a timeout that produces a specific set play out of the stoppage. If we write $Y_i(1)$ without specifying what version of a timeout we mean, the potential outcome is underspecified.

Sometimes this is acceptable because the treatment is a policy bundle: "the coach calls timeout using the team's normal timeout process." But then the causal effect is the effect of that bundle, not the effect of one specific tactical instruction.

7.7.2 Assumption 2: Exchangeability / No Unmeasured Confounding

Definition 7.5 (Conditional Exchangeability)**.** *Treatment assignment is conditionally independent of the potential outcomes given pre-treatment covariates X:*

$$(Y_i(1), Y_i(0)) \perp\!\!\!\perp Z_i \mid X_i.$$

This means that, after conditioning on X, treated and control units are comparable in the counterfactual sense. Among timeout opportunities with the same covariates, teams that called timeout would have had the same potential-outcome distribution as teams that did not, had both groups been assigned to the same decision. This is why the assumption is often called *no unmeasured confounding*: after adjusting for X, we are assuming there is no leftover hidden variable that still drives both treatment assignment and the outcome.

For the timeout example, X should include only pre-decision variables, such as:

- recent score swing over the previous 180 seconds;
- current score margin;
- time remaining;
- quarter and period;

- home/away;
- team and opponent strength entering the game;
- timeouts remaining, if available;
- lineup quality, fatigue, foul trouble, and injuries known before the decision, if available;
- coach or team tendencies measured before the decision.

This is often the most important and least testable assumption in sports studies. We can check whether observed covariates are balanced. We cannot directly check whether unobserved covariates are balanced.

Example 7.6 (Private Information)**.** Suppose a coach sees that the point guard is gassed, or notices a defensive coverage breakdown that the public play-by-play data does not record. That information may increase both the probability of calling timeout and the risk that the score swing continues. If the analyst does not observe that tactical information, timeout decisions may remain confounded even after adjusting for recent score swing, score margin, and time remaining.

7.7.3 Assumption 3: Positivity / Overlap

Definition 7.7 (Positivity)**.** *For every covariate profile x in the target population, each treatment level has positive probability:*

$$0 < \mathbb{P}(Z_i = 1 \mid X_i = x) < 1.$$

Positivity means that, for the kinds of situations in the population we care about, both treatment options actually occur with positive probability. In plain language, we need some situations where coaches call timeout and some similar situations where coaches do not; otherwise the data cannot support that comparison.

In timeout data, overlap is plausible in some regions and implausible in others. A -7 swing over the previous 180 seconds in the second or third quarter may be useful because some coaches call timeout and some do not. A much larger collapse, or a very late-game threshold crossing, may be less informative because almost everyone calls timeout. Causal claims should therefore be restricted to the part of the data where both treatment options were realistically available.

7.7.4 Assumption 4: Temporal Ordering and No Bad Controls

Adjustment variables should be measured before treatment. Adjusting for variables affected by the treatment can distort the causal comparison.

For timeout decisions, do not adjust for the lineup after the timeout, the score change over the next few possessions, or any post-timeout shot-quality metric. These variables are consequences of the timeout decision. Controlling for them can remove part of the effect we are trying to estimate. More controls are not automatically better; we want common causes of treatment and outcome, not consequences of the treatment.

7.7.5 Assumption 5: No Interference

Definition 7.8 (No Interference)**.** *Unit i's potential outcome depends only on unit i's own treatment, not on the treatments assigned to other units.*

A violation would occur if the treatment assigned to one unit changed the outcome for a different unit. Basketball makes this tricky because the game is sequential: one timeout changes lineups, rest, tactical choices, and the next set of possessions. Those downstream consequences are not a violation of the causal question themselves (they are part of the treatment effect for that timeout), but the practical concern is that, if we treat many timeout opportunities from the same game as separate units, one earlier decision may affect the context and outcome of a later one. We therefore need to define the unit carefully and account for this dependence when estimating uncertainty.

7.8 Core Methods for the Timeout Example

For this timeout example, the most natural tools are methods that try to make timeout and no-timeout situations more comparable before comparing outcomes.

Method	Basic idea	Timeout interpretation	Main weakness
Regression adjustment	Model $\mathbb{E}[Y \mid Z, X]$ and compare predicted outcomes under $Z = 1$ versus $Z = 0$.	Estimate the effect of calling timeout after controlling for recent score swing, score margin, time remaining, and team context.	Misspecified functional form or omitted confounders can turn the adjusted comparison into a misleading association.
Matching	Pair or group treated units with similar control units based on pre-treatment covariates.	Match timeout moments to no-timeout moments from similar swing states.	Poor matches leave treated and control moments systematically different, so the comparison remains confounded.
Propensity scores	Model $\mathbb{P}(Z = 1 \mid X)$ to summarize how likely a coach was to call timeout in that situation.	Estimate how timeout-prone the decision was relative to the observed context.	Missing strategic variables in the decision model can leave imbalance even after adjustment.
Weighting	Reweight moments so the timeout and no-timeout groups have more similar pre-treatment covariate distributions.	Create a pseudo-population in which observed swing states are more balanced across decisions.	Poor overlap creates extreme weights, so a small number of moments can dominate the estimate.
Sensitivity analysis	Ask how strong an unmeasured confounder would need to be to change the conclusion.	After estimating a timeout effect, quantify how much hidden tactical or fatigue information would be needed to explain it away.	It does not remove hidden bias; it quantifies vulnerability under a chosen sensitivity model, so misspecification can mislead.

Table 7.2: Core causal methods that map naturally onto the timeout example.

7.8.1 Regression Adjustment

The most familiar approach is regression adjustment. Fit an outcome model

$$\mathbb{E}[Y_i \mid Z_i, X_i] = m(Z_i, X_i),$$

and estimate

$$\widehat{\text{ATE}}_{\text{reg}} = \frac{1}{n}\sum_{i=1}^{n}\left\{\widehat{m}(1, X_i) - \widehat{m}(0, X_i)\right\}.$$

This formula clarifies what regression is doing causally. It predicts each unit's outcome under treatment and under control, then averages the predicted contrast.

A coefficient on Z in a regression is not automatically the effect of calling timeout. It is an adjusted association unless the design and assumptions justify reading it causally.

7.8.2 Matching, Propensity Scores, and Weighting

The propensity score is

$$e(X_i) = \mathbb{P}(Z_i = 1 \mid X_i).$$

It models treatment assignment, not the outcome. For each moment, it summarizes how likely a coach was to call timeout given the observed pre-timeout covariates. The purpose is balance: after matching or weighting on this information, timeout and no-timeout moments should look similar in their pre-timeout covariates.

Matching asks for direct comparisons between moments that looked similar before the decision. Weighting asks each moment to count more or less so that the treated and control groups become more comparable overall. The reason the propensity score is useful is that it compresses many observed covariates into one summary of how treatment was assigned, which can make balance checks and adjustments easier to organize.

A common inverse-probability weighted estimator is

$$\widehat{\text{ATE}}_{\text{IPW}} = \frac{1}{n}\sum_{i=1}^{n}\left\{\frac{Z_iY_i}{\widehat{e}(X_i)} - \frac{(1-Z_i)Y_i}{1-\widehat{e}(X_i)}\right\}.$$

The major practical danger is weak overlap. If some estimated propensity scores are near 0 or 1, the corresponding weights become large and the analysis depends heavily on a small number of moments. In timeout terms, we should be skeptical when the model tries to learn from situations where almost every coach calls timeout or almost none do.

7.8.3 Sensitivity Analysis

Even after adjusting for observed covariates, we may still worry about hidden information: a coach's private tactical concern, a player's fatigue, or a matchup problem not captured well in the data. Sensitivity analysis asks how strong that missing confounding would need to be to overturn the conclusion. In other words, it asks how large the hidden bias would have to be before we would stop believing the result.

A sensitivity model is simply a model for the hidden bias we are worried about. It specifies how an unmeasured confounder might affect treatment assignment and the outcome, and then asks how strong that confounding would need to be to materially change the estimate.

Sensitivity analysis does not remove bias. It tells us how fragile the conclusion is to the kinds of missing strategic information that are plausible in timeout decisions.

7.9 A Preview of Broader Causal Designs

The timeout example is especially useful for regression adjustment, matching, weighting, and propensity-score thinking. Other causal designs are easier to see in different sports settings, so it helps to briefly zoom out to the broader toolbox.

- **Difference-in-differences:** compare changes over time in a treated group to changes over time in a comparison group. The key assumption is parallel trends: absent treatment, the groups would have continued on similar paths.
- **Regression discontinuity:** exploit a sharp cutoff that changes treatment probability and compare units just above and below the cutoff. The key assumption is local continuity: near the cutoff, those units are otherwise comparable.
- **Instrumental variables:** use an outside source of variation that shifts treatment but affects the outcome only through that treatment. The key assumption is exclusion: the instrument changes the outcome only by changing treatment.

These can be powerful designs, but each one relies on its own identifying assumptions and needs to be argued for on its own terms.

7.10 Strengths and Weaknesses of Causal Inference

Causal inference helps because it makes the argument more explicit.

- **Strength: it clarifies the question.** We must say what intervention, comparison, outcome, unit, and time ordering we mean.
- **Strength: it separates question from method.** The causal estimand comes first; regression, matching, and weighting are only tools for approximating it.
- **Strength: it pushes us toward design.** Instead of treating modeling as automatic, it asks what comparison could make the treated and control units more alike.
- **Strength: it gives us diagnostics.** We can ask about balance, overlap, robustness, and sensitivity rather than treating one fitted model as the end of the story.

It also has real limits.

- **Weakness: the key assumptions are often untestable.** If the important confounders were not measured, the design cannot guarantee a valid causal estimate.
- **Weakness: more data do not solve hidden bias.** A very large sample can estimate the wrong comparison very precisely.
- **Weakness: prediction can look better than causation.** Flexible models may fit outcomes well while still failing to recover an intervention effect.
- **Weakness: sports data often violate clean-design ideals.** Dependence across observations, bundled treatments, and limited overlap can make the target comparison hard to approximate.

The right attitude is causal humility. The phrase "X causes Y" does *not* come for free. A fitted model, even with a small p-value and strong predictive accuracy, *is not causal* by itself. Causal results should be explained in terms of the comparison being made, the assumptions doing the work, and the limits of the design.

7.11 A Short Checklist

Before fitting a causal model, it is useful to work through a short checklist.

1. **State the intervention clearly.** What is being changed, relative to what alternative?
2. **Define the unit, time zero, outcome, and estimand.** What exactly is the comparison, and for whom?
3. **List the pre-treatment variables that matter.** Which covariates help determine both the decision and the outcome?
4. **Exclude bad controls.** Do not adjust for variables measured after treatment or affected by it.
5. **Ask whether a credible comparison is available.** Is there overlap, and can treated and control units be made reasonably comparable?
6. **Choose a design and estimator that match the problem.** Then check uncertainty, robustness, and sensitivity before using causal language.

Suggested Readings

[Holland 1986] Holland, P. W. (1986). Statistics and causal inference. *Journal of the American Statistical Association.*

[Imbens and Rubin 2015] Imbens, G. W., and Rubin, D. B. (2015). *Causal Inference for Statistics, Social, and Biomedical Sciences.* Cambridge University Press.

[Rosenbaum 2002] Rosenbaum, P. R. (2002). *Observational Studies.* Springer.

[Yu and Small] Yu, R., and Small, D. *Causal Inference from Observational Studies.*